



SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE CODE : CSA40

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COURSE NAME : MANAGEMENT INFORMATION SYSTEM

Reg NO: 192110204

10) Draw a use case diagram for Library Management System is an automation system used to manage a library and the different resource management required in it like cataloguing of books, allowing check out and return of books, invoicing, user management, etc. The user can search for books details using few book properties (Book ID, Title, Author, and Publisher). Searching should return details about all the book copies that match the search query.

Aim:

To design an **Online Security Management System** using a **Class Diagram**, which automates the **duty assignment, leave tracking, and payroll calculation** for security personnel within the college campus.

Procedure:

1. User Login & Authentication

- Library members and staff log in to the system with valid credentials.

2. Book Catalog Management

- Librarians add, update, or remove books from the catalog with details like Book ID, Title, Author, and Publisher.

3. Book Search & Lookup

- Users can search for books using Book ID, Title, Author, or Publisher to find available copies.

4. Book Checkout Process

- Registered users can borrow books, and the system updates their account with the due date.

5. Book Return & Fine Calculation

- Users return books, and fines are calculated automatically if the due date is exceeded.

6. User Management

- Librarians manage user registrations, membership renewals, and access permissions.

7. Invoice & Payment Handling

- The system generates invoices for overdue books, lost books, or membership fees.

8. Report Generation

- The system provides reports on book availability, borrowing history, and overdue books.

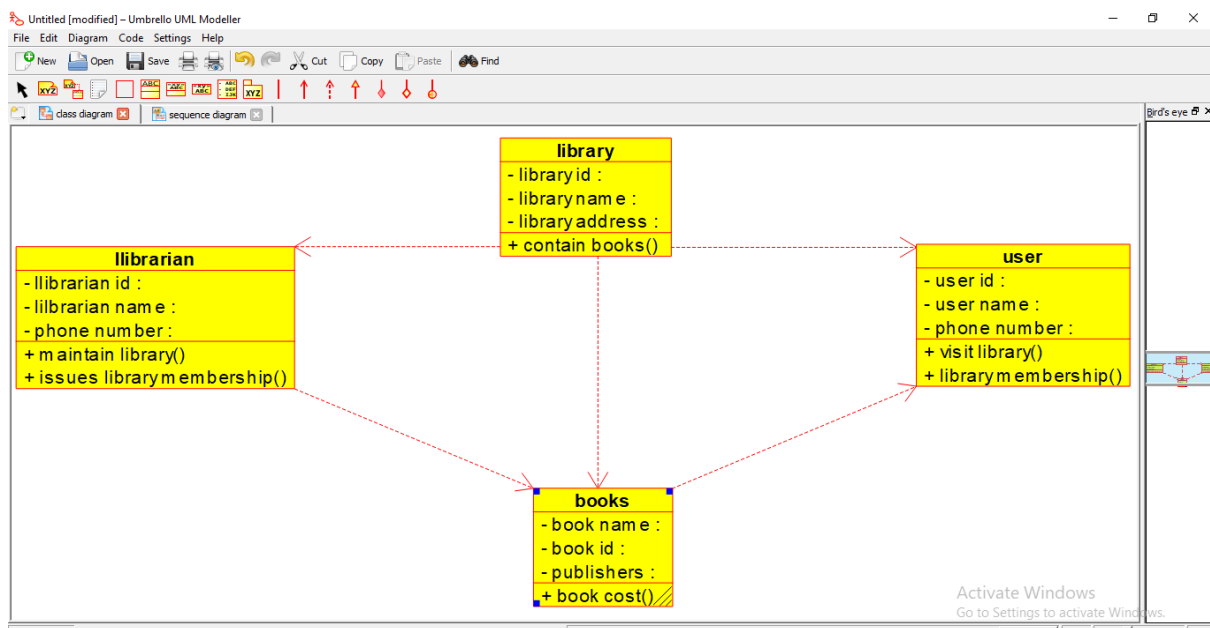
9. Reserve Books (Optional)

- Users can place a hold on books currently checked out to borrow them later.

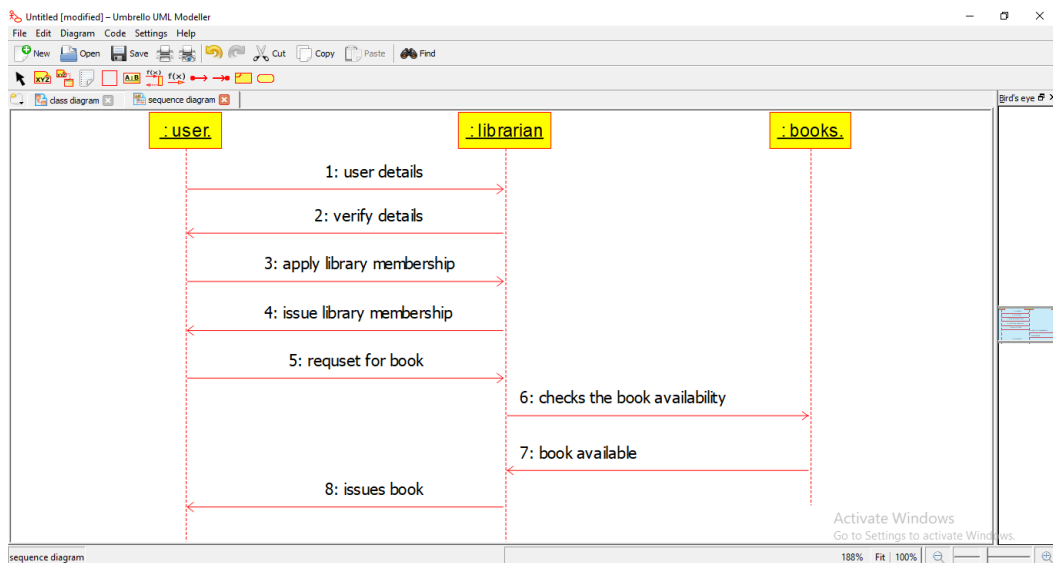
10. System Logout & Maintenance

- Users log out, and librarians ensure the system runs smoothly.

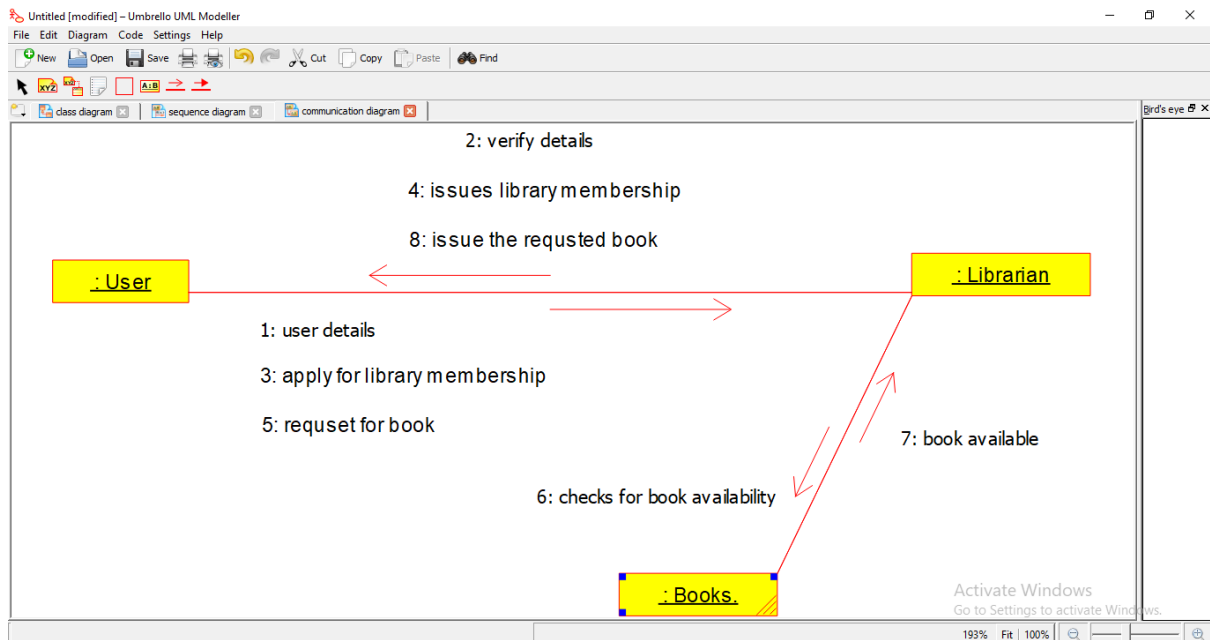
Class Diagram...



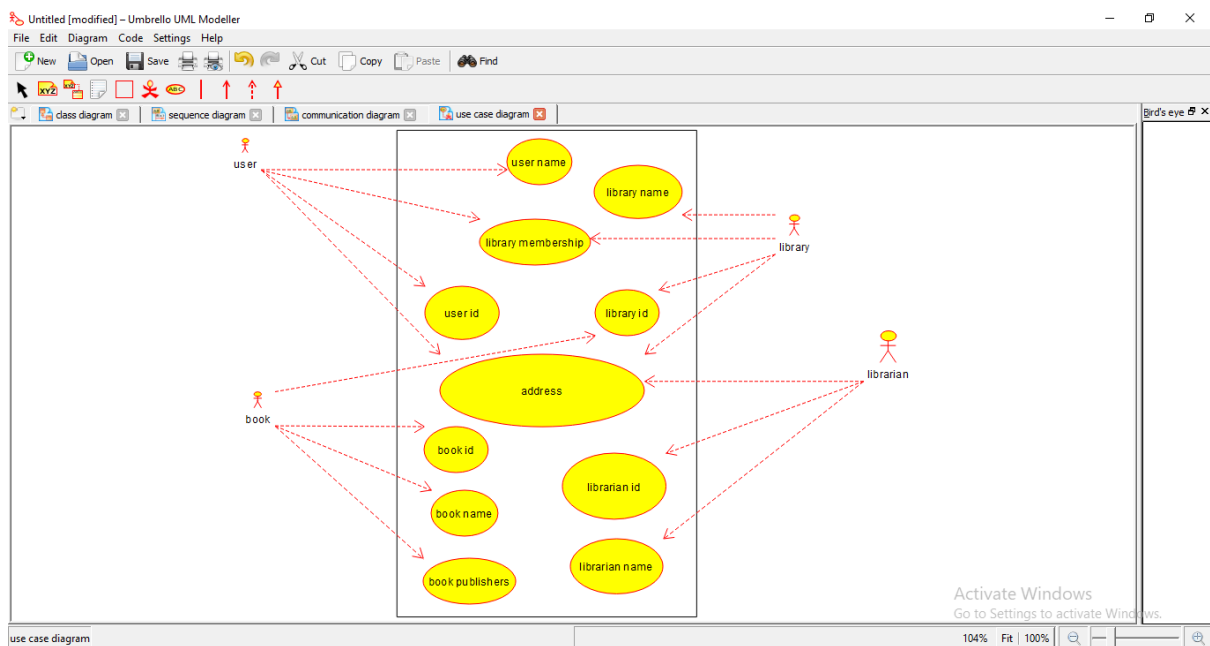
Sequence Diagram...



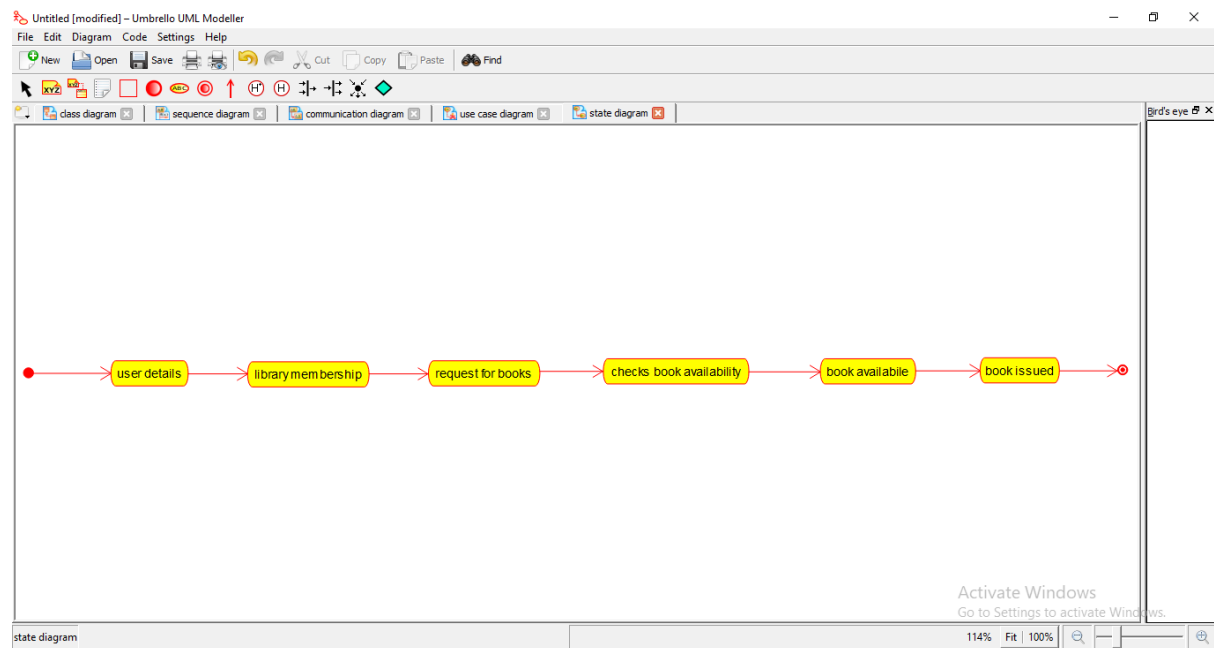
Communication Diagram...



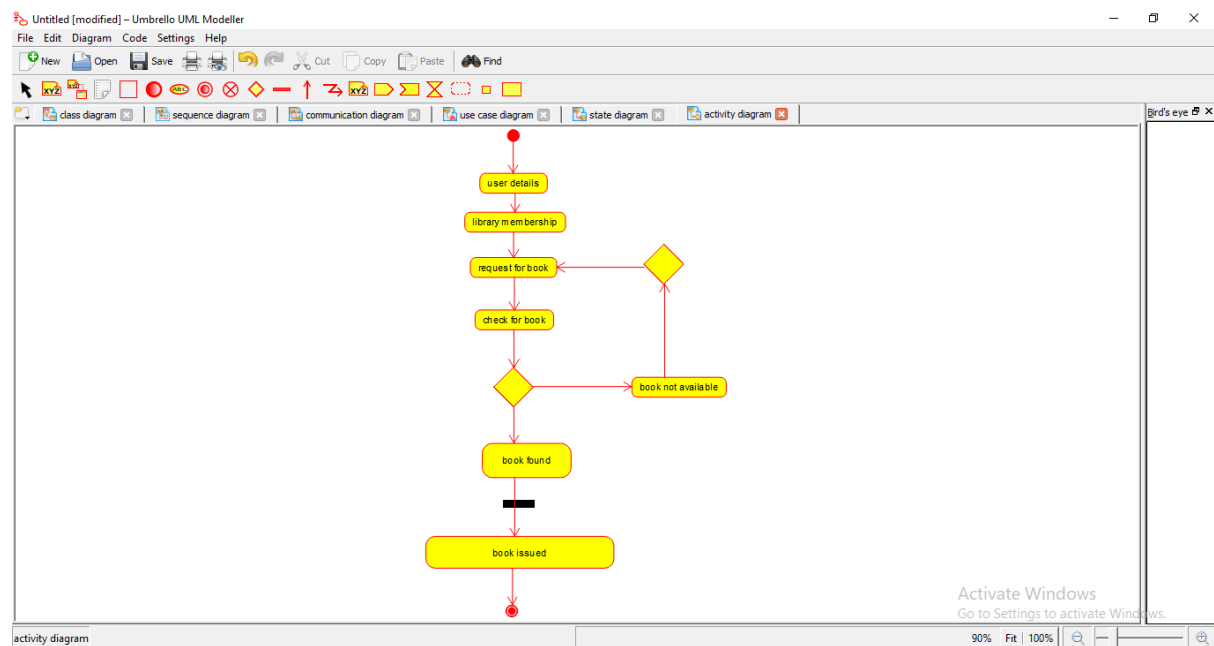
Use Case Diagram...



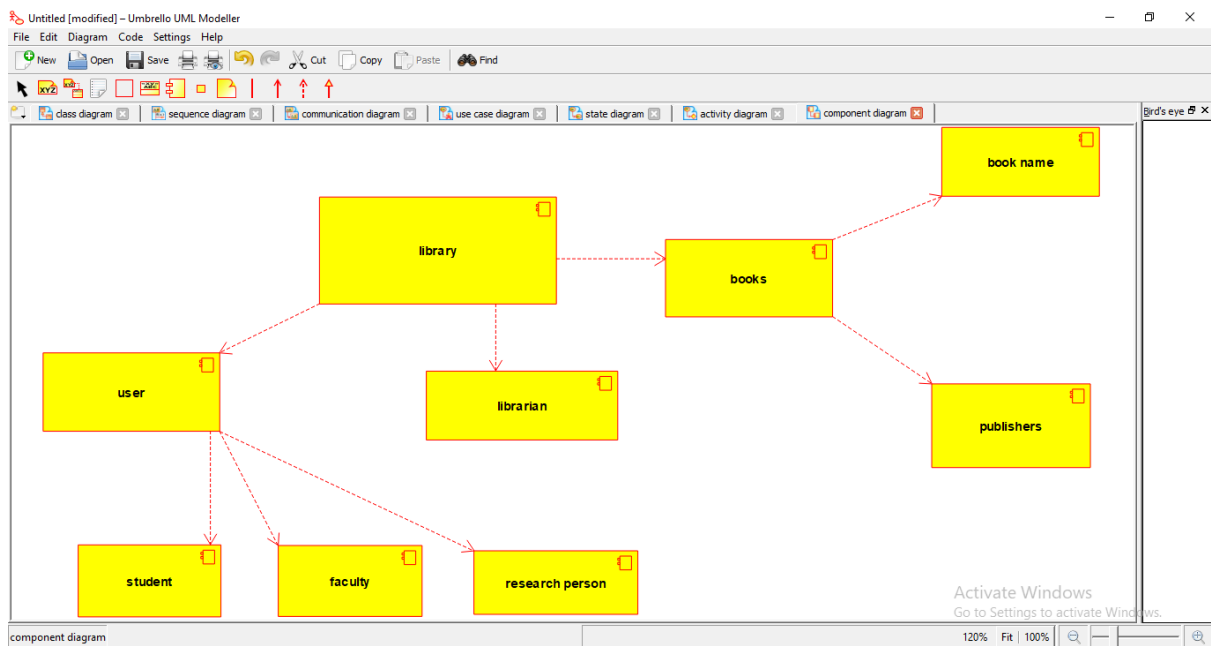
State Diagram...



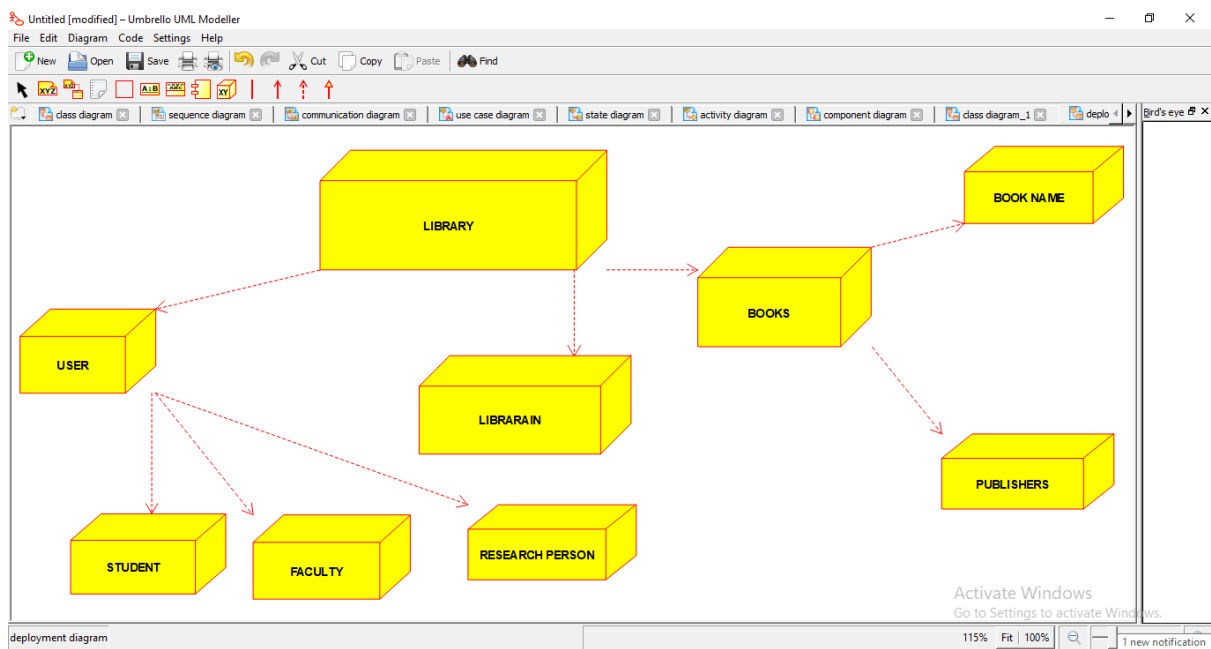
Activity Diagram...



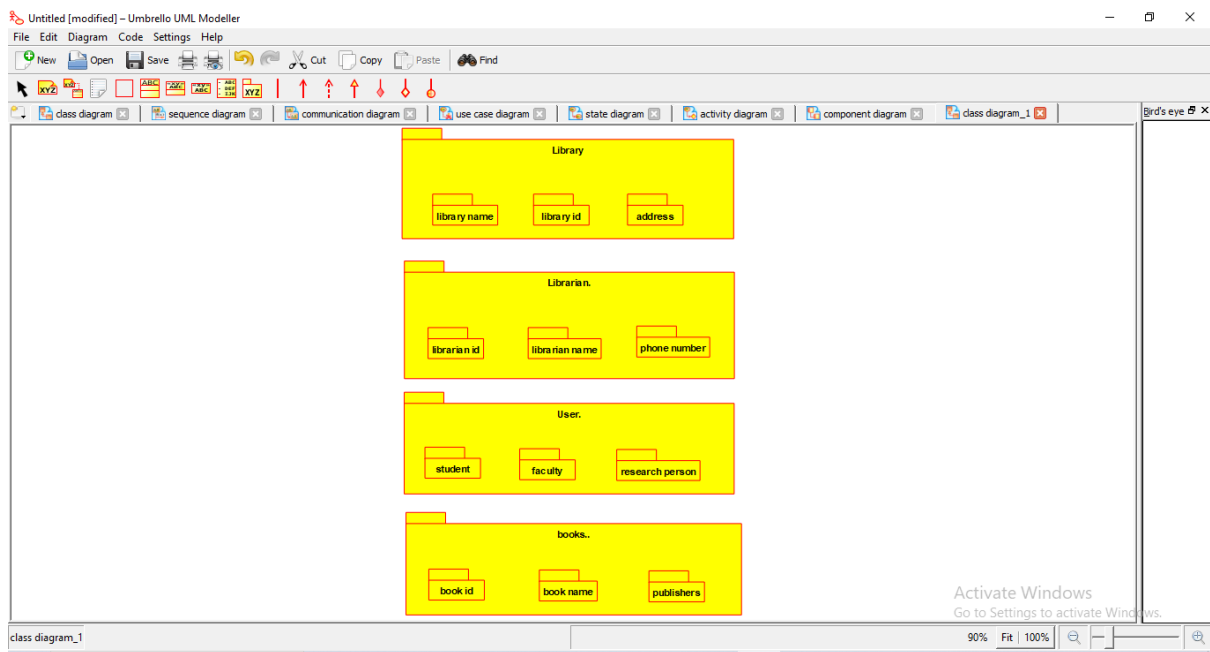
Component Diagram...



Deployment Diagram...



Entity Relationship Diagram...



Result:

The Library Management System Use Case Diagram effectively illustrates book cataloging, user interactions, borrowing, and invoicing processes. This system enhances library efficiency, improves book accessibility, and reduces manual workload.